

NELIREF NLP, LLM & Generative AI - Task 2 Assignment

Building Foundational Understanding of NLP, LLMs, and Generative AI

Course: NELIREF 3rd Data Science & AI Summer School 2026

Assignment: Introduction to NLP, LLM, and Generative AI - Task 2

Difficulty Level: Beginner/Introductory (concepts only, no heavy data processing)

Due Date: Wednesday, July 15, 2026 at 11:59 PM Eastern Daylight Time (EDT)

Submission: Email your repository link to newliferesearchconsult@gmail.com

Instructor: Joseph Luper Tsenum

Assignment Overview

This assignment builds on Task 1's discipline of version control and AI-assisted (vibe) coding, and now applies it to core concepts in Natural Language Processing (NLP), Large Language Models (LLMs), and Generative AI. You will:

- Review the instructor's concept notebook covering AI, ML, NLP, embeddings, attention, and LLMs
- Practice core NLP building blocks: tokenization, preprocessing, bag-of-words, and simple language models
- Use AI assistance (Copilot, Claude, or ChatGPT) to understand and extend starter code
- Complete 5 hands-on practice exercises in a dedicated notebook
- Document your learning journey with clear comments and git commits

Key Philosophy: This is not about memorizing definitions - it's about building intuition for how NLP, LLMs, and Generative AI actually work under the hood, using small, hand-built examples before you ever touch a real pretrained model.

Learning Objectives

By completing this assignment, you will:

1. **Understand core AI/ML concepts** - the difference between traditional programming and machine learning, and the three learning paradigms
2. **Understand the NLP pipeline** - tokenization, preprocessing, text representation, and embeddings
3. **Understand Transformers and attention** - why they replaced RNNs/LSTMs and how self-attention works conceptually
4. **Understand LLMs and Generative AI** - how language models predict text and how generative models create new content
5. **Practice AI-assisted coding** - use Copilot/Claude/ChatGPT effectively to build and debug small NLP building blocks
6. **Build documentation habits** - comment your code and your commits so your learning is traceable

Prerequisites

Before starting, ensure you have:

- **A GitHub account** (free at github.com)

- **Completion of Task 1** - familiarity with Codespaces, git, and AI-assisted coding
- **A Python environment** (Codespace, local machine, or cloud IDE such as Google Colab, Anaconda, etc.)
- **Familiarity with Jupyter Notebooks**

Don't have these? Check the Resources section at the end for setup guides, revisit Task 1's repository for environment setup instructions, or revisit the Roadmap to Data Science and AI where Pre-Lab session was incorporated.

Task Breakdown

PART 1: Repository Setup (Due: Wednesday, July 15, 2026)

Step 1a: Fork, Clone, or Download the Repository

Repository: <https://github.com/newliferesearchfoundation/Intro-to-NLP-LLM-and-Generative-AI>

Option A: Fork the Repository (Recommended for Learning)

```
# Go to https://github.com/newliferesearchfoundation/Intro-to-NLP-LLM-and-Generative-AI
# Click the "Fork" button in the top-right
# This creates YOUR OWN copy of the repository
```

Option B: Clone the Repository

```
git clone https://github.com/newliferesearchfoundation/Intro-to-NLP-LLM-and-Generative-AI.git
cd Intro-to-NLP-LLM-and-Generative-AI
git remote add origin https://github.com/YOUR-USERNAME/Intro-to-NLP-LLM-and-Generative-AI.git
git push -u origin main
```

Option C: Download & Create New Repository

```
# Download the ZIP file from the repository's "Code" button
# Extract it
# Create a new repository on GitHub
# Upload the content
# git clone your-new-repo
```

Step 1b: Create Your GitHub Codespace (or reuse your Task 1 environment)

1. Go to your forked repository on GitHub
2. Click the "Code" button (green button, top-right)
3. Select the "Codespaces" tab
4. Click "Create codespace on main"
5. Wait for the Codespace to initialize (2-3 minutes)

PART 2: Review the Concept Notebook (Due: Wednesday, July 15, 2026)

Task 2a: Run NELIREF_Basics_NLP_LLM_GenAI.ipynb

1. Open the notebook in your Codespace and select the Python kernel
2. Run every cell (Shift + Enter) in order, from Part 1 (What is AI?) through Part 8 (Summary & Key Takeaways)
3. For each Part, pause and answer in a new markdown cell: *"What is the one thing I want to remember from this section?"*
4. If a concept is unclear, ask your AI assistant to explain it with a different example before moving on

Sections covered: AI/ML foundations, types of learning, NLP basics, word embeddings, Transformers & attention, language models & LLMs, Generative AI, and a final summary with references.

PART 3: Practice Exercises (Due: Wednesday, July 15, 2026)

Task 3a: Complete NLP_LLM_GenAI_Practice_Exercises.ipynb

Open the companion practice notebook and complete all 5 exercises:

- Exercise 1: Build a Tokenizer
- Exercise 2: Bag-of-Words Word Counter
- Exercise 3: Text Preprocessing Pipeline (stopword removal)
- Exercise 4: Cosine Similarity for Word Embeddings
- Exercise 5: Mini Bigram Language Model (next-word predictor)

How to Approach These:

1. Read the exercise description and the docstring/TODO comments first
2. Try solving it yourself (10-15 minutes) before asking for help
3. If stuck, use Copilot/AI: ask *"How can I approach this?"*, *"Show me a similar example"*, or *"What's wrong with my code?"*
4. Run the provided test/assert cell under each exercise to self-check your answer
5. Add a comment explaining what you learned, for example:

```
# Learned: split() + strip(string.punctuation) is enough for simple tokenization
# Copilot helped me understand why lowercasing prevents duplicate tokens
def tokenize(text):
    return [t.lower().strip(string.punctuation) for t in text.split()]
```

PART 4: BONUS Challenge (Optional, Due: Wednesday, July 15, 2026)

Build a mini rule-based chatbot or attention visualizer!

Requirements:

- Create a new notebook: my_nlp_project.ipynb
- Build EITHER a simple keyword-matching chatbot OR a text-based attention-weight visualizer using the concepts from this assignment
- Use Copilot/Claude to help with planning, logic, and structure
- Document your code with comments explaining your design choices

Project Ideas:

- Rule-based FAQ chatbot (keyword → response mapping)
- Sentiment classifier using a hand-built positive/negative word list
- Attention-weight bar chart (text-based) for a sentence of your choice
- Simple n-gram "autocomplete" that suggests the next 3 words as you type

Example Structure:

```
# Project: Mini FAQ Chatbot
# Created with Copilot assistance
# Learned: dictionary lookups, keyword matching, string normalization

faq = {
    "hours": "We are open 9am - 5pm EDT, Monday to Friday.",
    "deadline": "The Task 2 deadline is July 15, 2026 at 11:59 PM EDT.",
```

```

}

def chatbot_reply(user_input):
    """Return a canned reply based on keyword matching"""
    text = user_input.lower()
    for keyword, reply in faq.items():
        if keyword in text:
            return reply
    return "I'm not sure - please ask an instructor!"

print(chatbot_reply("What is the deadline?"))

```

PART 5: Version Control & Submission

Step 5a: Commit Your Changes

```

# In your Codespace terminal, run:

# 1. Check what files changed
git status

# 2. Add all changes
git add .

# 3. Commit with a meaningful message
git commit -m "Task 2: Completed NLP/LLM/GenAI concepts and exercises

- Reviewed NELIREF_Basics_NLP_LLM_GenAI.ipynb and added reflection notes
- Completed all 5 exercises in NLP_LLM_GenAI_Practice_Exercises.ipynb
- Added comments documenting Copilot-assisted learning
- [Optional] Created bonus project: my_nlp_project.ipynb"

# 4. Push to GitHub
git push origin main

```

Step 5b: Verify Your Changes

1. Go to your GitHub repository
2. Refresh the page to see your changes
3. Click on commits to verify your commit message
4. Open the notebooks to confirm changes are saved

Step 5c: Submit link to your GitHub repository

Email: newliferesearchconsult@gmail.com

Subject Line:

```
NELIREF Task 2 Submission - [YOUR NAME] - NLP, LLM & Generative AI
```

Email Body:

```

Hello,

Please find my Task 2 submission below:

GitHub Repository: https://github.com/YOUR-USERNAME/Intro-to-NLP-LLM-and-Generative-AI

```

Completed:

1. Forked/cloned Intro-to-NLP-LLM-and-Generative-AI repository
2. Created GitHub Codespace (or used local/Colab environment)
3. Ran and reflected on NELIREF_Basics_NLP_LLM_GenAI.ipynb
4. Completed all 5 exercises in NLP_LLM_GenAI_Practice_Exercises.ipynb
5. [Optional] Created bonus project: my_nlp_project.ipynb
6. Committed and pushed all changes to main branch

AI Tools Used: [Copilot / Claude / ChatGPT / ChatGPT-4o]

Platform: [GitHub Codespace / VS Code Local / macOS / Other]

Thank you,
[YOUR NAME]

Grading Rubric

Criteria	Points	Requirements
Repository Setup	10	Fork/clone complete, Codespace (or local/Colab) environment accessible
Concept Notebook Review	15	NELIREF_Basics_NLP_LLM_GenAI.ipynb run fully, all concept cells studied and understood
Practice Exercises	40	All 5 exercises in NLP_LLM_GenAI_Practice_Exercises.ipynb completed correctly with comments
Concept Quiz (in-notebook)	10	Short reflection/quiz cells answered thoughtfully
Code Documentation	15	Clear comments explaining changes, choices, and what was learned
Git Workflow	10	Proper commits, meaningful messages, pushed to main
Bonus Challenge	20	Optional: +20 bonus points for the mini chatbot / attention visualizer
TOTAL	100	(Plus 20 bonus)

Troubleshooting Common Issues

"Copilot is not working in my Codespace"

- Make sure you're signed into GitHub
- Restart the Codespace
- Try reinstalling the GitHub Copilot extension

"My notebook has a kernel error"

```
# In terminal: pip install --upgrade jupyter ipykernel python -m ipykernel install --user
```

"My cosine similarity/bigram model exercise gives NaN or a KeyError"

- Check that you are not dividing by zero (a zero-length vector has no direction)

- Check that the word you are looking up actually exists in your dictionary - print the dictionary keys to confirm
- Ask your AI assistant: "Why am I getting a KeyError here?" and paste the exact traceback

"Git push is being rejected"

```
# Make sure you're on main branch git branch # If not on main: git checkout main git
pull origin main git push origin main
```

"I don't understand a concept (e.g. attention, embeddings)"

- Re-read the relevant Part in NELIREF_Basics_NLP_LLM_GenAI.ipynb
- Ask Copilot/Claude/ChatGPT: "Explain [concept] using a completely different everyday example"
- Ask NELIREF Summer School Instructors and TA on Telegram at <https://t.me/neliref> or visit our website at www.neliref.org

Resources & Support

GitHub & Git Guides

- GitHub Codespaces Documentation - <https://docs.github.com/en/codespaces>
- Git Basics (Official Git Book) - <https://git-scm.com/book/en/v2/Getting-Started-Git-Basics>

NLP / LLM / Generative AI Resources

- Vaswani et al. (2017) - "Attention Is All You Need"
- Devlin et al. (2019) - "BERT: Pre-training of Deep Bidirectional Transformers"
- Brown et al. (2020) - "Language Models are Few-Shot Learners" (GPT-3)
- Stanford CS224N: NLP with Deep Learning (course materials, free online)
- Hugging Face NLP Course - <https://huggingface.co/learn/nlp-course>

AI Assistance & Vibe Coding

- GitHub Copilot Documentation - <https://docs.github.com/en/copilot>
- Using Claude for Code - <https://claude.ai> - Free tier available
- ChatGPT for Beginners - <https://openai.com/chatgpt>
- **Vibe Coding Philosophy:** Use AI as a learning partner - you remain the boss.

Getting Help

- Ask Copilot/Claude: "I'm getting X error, how do I fix it?"
- Check Stack Overflow: search your exact error message
- NELIREF Social Media/Chat: join community platforms at www.neliref.org
- Email instructors and Teaching Assistants: newliferesearchconsult@gmail.com

Tips for Success

Best Practices

1. Code Daily - even 30 minutes a day beats 6 hours once a week
2. Understand Before Copying - read Copilot's suggestions before accepting them
3. Comment Everything - future you will thank present you

4. Use Git Commits - commit every time you finish a section, not just at the end
5. Test Your Code - re-run the assert/test cells to confirm your exercises pass

Using AI Assistants Effectively

DO:

- Ask specific questions: "Why does cosine similarity return a negative number here?"
- Request explanations: "Explain what self-attention does in one paragraph"
- Ask for help understanding: "What does a bigram model actually store?"
- Request similar examples: "Show me another way to remove stopwords"

DON'T:

- Just copy-paste code without reading it
- Skip understanding - an explanation you don't understand isn't helpful
- Use AI as a replacement for thinking - use it to think together

Building the Discipline

The goal of this assignment is to help you:

- Think like an NLP practitioner - breaking language problems into small, testable steps
- Learn from errors - understanding what went wrong and why
- Use modern tools - AI is standard in 2026 programming
- Document your work - writing comments and commit messages
- Version your code - saving your progress safely

Success Looks Like

By Wednesday, July 15 at 11:59 PM EDT, you should have:

- A GitHub repository with your forked/cloned code
- Multiple git commits showing your progress
- NELIREF_Basics_NLP_LLM_GenAI.ipynb run fully with your reflection notes added
- All 5 exercises completed in NLP_LLM_GenAI_Practice_Exercises.ipynb
- Comments in your code explaining what you learned
- [Bonus] A mini NLP project notebook (optional)
- Your repository link submitted via email

Frequently Asked Questions

Q: Can I use a different AI tool instead of Copilot?

A: Yes! Claude, ChatGPT, and ChatGPT-4o all work great. The skill being practiced is the same.

Q: Do I need to install any NLP libraries (spaCy, NLTK, transformers)?

A: No. This assignment intentionally uses only core Python (strings, dictionaries, lists, math) so you understand the mechanics before relying on libraries.

Q: Do I have to complete the bonus project?

A: No, it's optional. But it's a great way to practice and have fun!

Q: What if I can't finish by July 15?

A: Submit what you have. Partial work is better than no submission. Email if you need accommodation. Changes pushed to this repository after the original deadline will also be accepted until July 28, 2026.

Q: Can I work with a partner?

A: You can discuss concepts, but each person must complete and submit their own repository.

Q: What if my similarity/prediction results don't match the expected output exactly?

A: Focus on understanding why. Email the instructors and teaching assistants if you're stuck.

Support Contact

For Questions or Issues:

- Email: newliferesearchconsult@gmail.com
- Subject: "NELIREF Task 2 - [Your Question]"

For Technology Help:

- GitHub Copilot Issues: check the GitHub Copilot status page
- Codespace Issues: see the GitHub status page
- Python/Notebook Issues: post to Stack Overflow with your exact error message

Timeline Summary

Date	Milestone	Status
June 19-27, 2026	Instruction Days (Summer School Sessions)	Complete
July 1, 2026	Assignment Released	Today
July 2-7, 2026	Work on Tasks	In Progress
July 15, 2026 (11:59 PM EDT)	DEADLINE – Git commits until July 28	Due
August 2026	Feedback Provided	After deadline

Final Thoughts

This assignment moves you from general programming discipline into the specific mental models used to build modern AI systems: tokenization, representation, attention, and generation. The same building blocks you code by hand here - counting words, comparing vectors, predicting the next token - scale up (with far more data and compute) into the LLMs and Generative AI tools you use every day.

You're not just learning NLP. You're learning how the language technology of 2026 actually works.

Let's get started!

Happy Coding!

*Created for NELIREF 3rd Data Science & AI Summer School 2026
Emphasizing Discipline, Vibe Coding, and Modern Development Practices*